A Novel Method for Hyperspectral Image Classification: Deep Network With Adaptive Graph Structure Integration

Bing Yang, Feilong Cao, and Hailiang Ye

***Abstract*— Hyperspectral image (HSI) classification has always been one of the hot issues in the study of geographic remote sensing information, and graph neural networks have attracted much attention in recent years. Several graph neural network-based approaches have been introduced into HSI study to explore the spatial information of HSI within a constructed graph. However, the existing methods of building HSI-based graphs are always unsuitable and inaccurate due to the complicated spatial variability of spectral signatures. Meanwhile, these graph-based HSI classification methods usually suffer from the over-smoothing problem. Motivated by these, this article presents a novel deep network with adaptive graph structure integration (DNAGSI), which could learn a graph structure of HSI dynamically and promote its discriminative ability with devising a much deeper network architecture. Specifically, dynamic graphs are first built across different layers and adaptively integrated with the initial graph structure to boost the robust graph representation of HSI. Second, the initial residual and identity mapping are employed to significantly increase the depth of the network and obtain more abstract deep features. Finally, a joint loss with center loss is devised to learn the similarity relationship between hyperspectral pixels explicitly, thereby gathering the intraclass graph features. Benefiting from the integration of center loss, initial residual, and identity mapping, the proposed method can alleviate the over-smoothing problem effectively to some extent. Experiments on benchmark HSI datasets demonstrate the superiority of DNAGSI over state-of-the-art methods.**

***Index Terms*— Deep learning, graph neural network, graph representation learning, hyperspectral image (HSI) classification.**

I. Introduction

ATTRIBUTED to the rapidly developing sensing technology, hyperspectral images (HSIs) can be easily collected with hundreds of contiguous narrow bands and detailed spatial distribution of land covers [1]–[3]. HSI classification is an important task, which distinguishes different physical materials with subtle spectral signatures benefiting from the abundant information and has received considerable attention in a wide range of applications, including urban planning, geological science, and environmental surveillance [4]–[6].

Manuscript received December 11, 2021; revised January 30, 2022 and February 7, 2022; accepted February 7, 2022. Date of publication February 9, 2022; date of current version March 23, 2022. This work was supported in part by the Zhejiang Provincial Natural Science Foundation of China under Grant LQ20F030016 and in part by the National Natural Science Foundation of China under Grant 62006215 and Grant 62176244. *(Corresponding author: Hailiang Ye.)*

The authors are with the College of Sciences, China Jiliang University, Hangzhou 310018, China (e-mail: bingyang0517@163.com; icteam@163.com; yhl575@163.com).

Digital Object Identifier 10.1109/TGRS.2022.3150349

There have been numerous works on HSI classification tasks over the last decade. The earlier works, such as support vector machine-based method [7], manifold ranking model [8], regularized local discriminant embedding approach [9], principal component analysis-based, and functional data-based techniques [10], [11], relied on handcrafted features. The classification performances of these works are likely to decay when the manual features are chosen inappropriately. Recently, due to the powerful representation capability of deep learning, HSI classification has been revolutionized by various deep models [12]–[15]. Especially, convolutional neural networks (CNNs) have shown great potential in feature extraction for HSI and, thus, facilitated subsequent classification tasks. For example, Yu *et al.* [16] presented an efficient CNN framework to boost the HSI classification performance. Lee and Kwon [17] built a multiscale-filter-based contextual CNN (CCNN) to explore local spatial relationships in HSI. Subsequently, Cao *et al* [18] incorporated the Markov random fields into CNN to solve the HSI classification tasks. Recently, Tang *et al.* [19] devised a 3-D octave convolution-based neural network to capture deep features of HSI. Sun *et al.* [20] established an end-to-end fully convolutional segmentation network for HSI classification. Gao *et al.* [21] introduced the mixed depthwise convolution into HSI classification, obtaining multiscale features and simultaneously reduce the computational cost. In addition, Zhong *et al.* [22] used the residual connection and constructed both spectral residual module and spatial residual module to exploit spectral and spatial information. Guo *et al.* [23] designed a novel feature-grouped network based on the attention mechanism to extract valuable spectral–spatial features. Among the above-mentioned methods, adjacent hyperspectral pixels are represented as a small data cube to utilize spatial correlation of homogeneous regions. However, these works still exist several issues. First, despite the fact that hyperspectral pixels inside a homogeneous area are more likely to belong to the same class, the cube inevitably contains many different-class hyperspectral pixels, especially at their boundary. Furthermore, while the cube can contain local spatial correlation information, previous CNN-based approaches use the implicit assumption that these cubes are independent of one another. It is usually unreasonable since the hyperspectral pixel is highly related to others in the same class. This suggests that exploring the abundant relationships and interdependence between hyperspectral pixels is still undervalued.

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:55:16 UTC from IEEE Xplore. Restrictions apply.

Actually, hyperspectral pixels of the same-class tend to have comparable hyperspectral signatures. Even if they are far away, each hyperspectral pixel should gain supplemental information from other same-class pixels. Graph convolutional networks (GCNs) have lately been used to better utilize complex interactions and interdependence between hyperspectral pixels. For instance, Mou *et al.* [24] built a nonlocal GCN to learn graph representations of HSI using the whole image as input. To improve classification performance and simultaneously reduce computational cost, Hong *et al.* [25] developed a minibatch GCN and combined it with CNN for HSI classification. Sha *et al.* [26] constructed a semisupervised framework for HSI classification based on graph attention networks. Qin *et al.* [27] created a spectral–spatial GCN to use HSIs' spectral and local spatial information simultaneously. To acquire more accurate results, He *et al.* [28] used GCN to extract deep features as well as label distribution from HSI. Most existing approaches, however, assume that the initial constructed graph structure is accurate and appropriate, and they use the fixed initial graph to aggregate the spatial neighborhood information. They are unable to capture the dynamic graph topology in each layer of GCN in an adaptive manner, limiting the GCN's representation capabilities and thereby affecting the performance of the subsequent task.

Recent research has demonstrated that investigating proper graph structure is critical to obtaining discriminative graph representations of HSI. For example, Wan *et al.* [29] attempted to capture the multiscale information of HSI by using a multiscale dynamic graph convolutional network. Again, Liu *et al.* [30] established a CNN-enhanced GCN (CEGCN), where the adjacent matrix of GCN was learned to some extent. Wan [31] tried to learn an improved similarity measurement to refine the connective links and edge weights in all graph convolutional layers, allowing the graphs to adapt to features in different layers. Ding *et al.* [32] proposed optimizing the initial graph by using graph topological consistent connectivity, leveraging global high-order neighbors adaptively in their model to improve HSI classification performance.

Nevertheless, most of these approaches typically ignore the initial graph of HSI, which is a potential weakness. In fact, though the initial graph cannot exactly represent the intrinsic similarities between the hyperspectral pixels, it certainly contains lots of essential information contained in the original HSIs. Accordingly, building a more refined graph structure is preferred for HSI graph representation extraction. Besides, all these approaches fall into the shallow GCN architectures and only employ a few graph convolutional layers (two layers in most works), inevitably restricting the performance. The main reason is that they cannot be generalized to much deeper architectures due to the over-smoothing problem [33]. Consequently, it is essential to relieve the negative effects while significantly increasing the model depth to extract high-level features from HSIs.

To overcome the above-mentioned limitations, this article presents a deep network with adaptive graph structure integration (DNAGSI) in an attempt to obtain discriminative graph representations of HSI and promote its classification performance. To begin, considering the fact that the initial graphs generated by many previous GCN-based approaches are not always dependable, dynamic graphs are first constructed and then adaptively integrated with the initial graph in different layers. This forms a novel adaptive graph structure integration module. The idea of adopting the dynamic graph can supply more available information to initial graph data and facilitate the stability of graph representation capabilities. Second, a joint loss is developed by introducing the center loss to regularize the learned graph representations of HSIs, which confine the graph representation to move as close to the matching class center as possible and thereby enhance their discriminative ability. Finally, to considerably improve model depth and relieve the over-smoothing problem, we incorporate the initial residual and identity mapping into this new framework. Compared with previous works, the crucial differences of the proposed approach lies in two aspects. On the one hand, during the graph structure learning process, the proposed approach captures the refined graph structure by adaptively integrating the information both in the initial graph and the dynamic graph at the node level to explore the complementary information in each other, while the existing works either neglect the importance of the initial graph or adopt some nonnode level integration approaches. On the other hand, the proposed DNAGSI significantly deepens the network depth and extracts more abstract and discriminative features from HSIs, whereas earlier works are generally shallow frameworks. The main contributions can be highlighted as follows.

1) A novel DNAGSI framework is developed, which constructs a dynamically generated graph structure in each layer and adaptively integrate these new graph structures of different layers with the initial graph structure to further refine the information. This can produce significantly more reliable and effective HSI graph representations.
2) The initial residual and identity mapping are fused in order to benefit DNAGSI from a much deeper network structure. As network depth increases, this could produce better high-level graph feature representations and meanwhile the over-smoothing phenomenon is alleviated to a certain extent.
3) A new joint loss is created by incorporating a center loss into the DNAGSI framework. This assures that the learned graph representations of intraclasses can cluster together, thereby facilitating the separability of intraclasses and interclasses.

The remainder of this work is structured as follows. Section II gives some notations and introduces the graph convolutional network. Section III discusses the proposed framework concretely. Section IV presents the experimental results and the corresponding analyses on real-HSI datasets. Section V contains the conclusions.

II. Preliminary

A. Notations

The goal of HSI classification task is to assign each hyperspectral pixel to a certain class. For the sake of clarity, some notations are defined before describing the proposed method.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:55:16 UTC from IEEE Xplore. Restrictions apply.

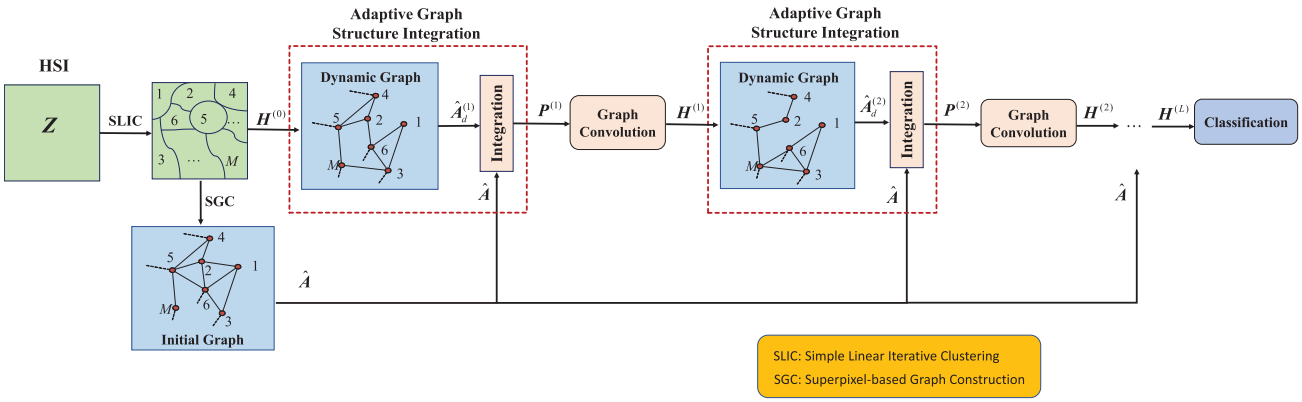


Fig. 1. Overview of the DNAGSI-based HSI classification framework. HSIs are first partitioned into numerous spatially related superpixels to construct an initial graph $\hat{\boldsymbol{A}}$. Then, dynamic graphs $\hat{\boldsymbol{A}}_d^{(l)}$ $(i = 1, 2, \ldots, L)$ are captured and integrated with the initial graph adaptively across multiple layers in order to obtain a refined graph $\boldsymbol{P}^{(l)}$. Finally, the discriminative graph representations $\boldsymbol{H}^{(l)}$ of HSI are learned by feeding and propagating the integrated graphs in different graph convolutional layers.

Let $\boldsymbol{Z} \in \mathbb{R}^{H\times W\times B}$ denote an HSI, where H, W, and B are, respectively, the height, the width, and the number of spectral bands. Assume that $\mathcal{C} = \{1, 2, \ldots, C\}$ (C is the number of classes) is the set of class labels for $\boldsymbol{Z}$. The samples set with labels is represented as $\mathcal{D}_L = \{(\boldsymbol{z}_{l_i}, y_i)\}_{i=1}^{N}$, where N denotes the number of labeled samples, and $\boldsymbol{z}_{l_i}$ and $y_i \in \mathcal{C}$, respectively, represent the spectral signature and class label of the ith labeled sample. In addition, $\|\cdot\|$ represents the Frobenius norm.

B. Graph Convolutional Network

Recently, graph convolutional networks have shown remarkable power in capturing complex interactions among data by aggregating neighborhood feature information via a message-passing mechanism [34]–[36]. Assume there are K nodes in a graph and f hidden units in the graph convolutional layer, then a widely utilized type of graph convolution, proposed by [37], is defined as

$$\boldsymbol{X}^{\text{out}} = \tilde{\boldsymbol{A}}^{*}\boldsymbol{X}^{\text{in}}\boldsymbol{W} \tag{1}$$

where $\tilde{\boldsymbol{A}}^{*} = \tilde{\boldsymbol{D}}^{-1/2}(\tilde{\boldsymbol{A}} + \boldsymbol{I}_K)\tilde{\boldsymbol{D}}^{-1/2} \in \mathbb{R}^{K\times K}$ is a normalized version of the adjacency matrix $\tilde{\boldsymbol{A}}$ of the input graph, $\boldsymbol{I}_K$ is the identity matrix, $\tilde{\boldsymbol{D}}$ is the degree matrix for $\tilde{\boldsymbol{A}}+\boldsymbol{I}_K$, $\boldsymbol{X}^{\text{in}} \in \mathbb{R}^{K\times d}$ represents the d-dimension features of K nodes in this graph, and $\boldsymbol{W} \in \mathbb{R}^{d\times f}$ is the learnable weight matrix.

III. Proposed Framework

In this section, the whole DNAGSI framework is proposed, as shown in Fig. 1. We aim to incorporate underlying dynamic graphs via a deep adaptive graph structure integration network while also benefiting from the center loss to reduce the impact of the over-smoothing problem and improve feature representation. Concretely, HSIs are first segmented into a large number of spatially related superpixels to form an initial graph. The dynamically updated graphs are then captured and integrated with the initial graph adaptively across multiple layers so as to produce a refined graph. Finally, the discriminative graph representations of HSI are learned by feeding and propagating the integrated graphs in different graph convolutional layers. Furthermore, to increase the network depth, the initial residual and identity mapping are introduced into graph convolutional layers. Meanwhile, during the training stage, a joint loss is devised by embedding the center loss to regularize the graph representations.

A. Superpixel-Based Graph Construction

In HSI classification, the GCN-based approaches typically work on superpixel-based graph but not the pixel-based graph. The reason is that if all hyperspectral pixels are considered as graph nodes, the computation cost could be prohibitively expensive. Therefore, our method also takes a superpixel-based graph as input. To this end, we convert the HSI into a superpixel-based graph. The detailed procedures are illustrated as follows.

To begin, the linear discriminant analysis (LDA) is employed for HSI dimensionality reduction to improve the efficiency of creating a graph structure. The HSI is then partitioned into numerous spatially related superpixels using the simple linear iterative clustering (SLIC) [38], denoted by $M = \lceil (H \times W)/\mu \rceil$ superpixels (here, $\mu \geq 1$ is the scale hyperparameter). As a result, HSIs can be transformed into a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ and $\mathcal{E}$ represent its corresponding node and edge sets, respectively. In this work, $\mathcal{V}$ is composed of all superpixels, and $\mathcal{E}$ denotes the connected edges. Each superpixel node $v_i \in \mathcal{V}$ is associated with a spectral signature $\boldsymbol{x}_i \in \mathbb{R}^B$, which is the average of the spectral signatures of all hyperspectral pixels in that superpixel. Therefore, the input feature of the graph is denoted by $\boldsymbol{X} \in \mathbb{R}^{M\times B}$. Let $\boldsymbol{A} \in \mathbb{R}^{M\times M}$ be the adjacency matrix of $\mathcal{G}$ and its element indicates the existence of an edge between each pair of superpixel nodes obtained by

$$A_{ij} = \begin{cases} \exp(-\nu\|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2), & \text{if } v_i \text{ and } v_j \text{ are adjacent,} \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

where ν is a hyperparameter and empirically set as 0.2 in this work. Note that nodes v_i and v_j are adjacent when they share a common boundary. The adjacency matrix $\boldsymbol{A}$ constructed in this manner clearly reveal the spatial structure of HSI. Obviously, the normalized adjacency matrix of initial graph is described as

$$\hat{\boldsymbol{A}} = \hat{\boldsymbol{D}}^{-\frac{1}{2}}(\boldsymbol{A} + \boldsymbol{I}_M)\hat{\boldsymbol{D}}^{-\frac{1}{2}} \in \mathbb{R}^{M\times M} \tag{3}$$

where $\hat{\boldsymbol{D}}$ is the degree matrix of $\boldsymbol{A} + \boldsymbol{I}_M$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:55:16 UTC from IEEE Xplore. Restrictions apply.

B. Deep Network With Adaptive Graph Structure Integration

In reality, the proposed method can be mainly separated into three phases. We will present them in the following in detail.

1) Dynamic Graph Structure Learning: The superpixel-based graph is widely used as the input of GCN in most superpixel-based approaches to obtain the graph representation of HSI. However, these superpixel-based graph methods always contain many unreliable edges due to the complicated spatial variability of spectral signatures, which may lead to negative impact during the feature learning. To overcome this issue, we propose in this article to build dynamic graphs layer by layer and merge them with the initial superpixel-based graph to provide a robust feature representation.

Actually, the dynamic graph is constructed using the cosine similarity between the hidden feature representations of each superpixel node pair. Specifically, for each node pair (v_i, v_j), we first calculate their cosine similarity as

$$S_{ij}^{(l)} = \frac{\left(\boldsymbol{h}_i^{(l)}\right)^{\mathrm{T}}\left(\boldsymbol{h}_j^{(l)}\right)}{\left\|\boldsymbol{h}_i^{(l)}\right\|\left\|\boldsymbol{h}_j^{(l)}\right\|} \tag{4}$$

where $\boldsymbol{h}_i^{(l)} \in \mathbb{R}^n$ denotes the hidden feature of node v_i in the lth ($l = 1, 2, \ldots, L$) graph convolutional layer, and n is the dimension of node features. Note that, in this article, the number of hidden units of all graph convolutional layers are all set to n.

Then, at the lth layer, the other nodes with the greatest k cosine similarity are chosen for each node to form the edges of the constructed dynamic graph. The adjacency matrix of the constructed dynamic graph in the lth layer is denoted as $\boldsymbol{A}_d^{(l)}$. The corresponding normalized adjacency matrix is

$$\hat{\boldsymbol{A}}_d^{(l)} = \left(\hat{\boldsymbol{D}}_d^{(l)}\right)^{-\frac{1}{2}}\left(\boldsymbol{A}_d^{(l)} + \boldsymbol{I}_M\right)\left(\hat{\boldsymbol{D}}_d^{(l)}\right)^{-\frac{1}{2}} \tag{5}$$

where $\hat{\boldsymbol{D}}_d^{(l)}$ is the degree matrix of $\boldsymbol{A}_d^{(l)} + \boldsymbol{I}_M$.

2) Adaptive Graph Structure Integration: A fusion module is required to merge the initial graph and the dynamic feature graph in order to successfully supplement the adjacency information of the dynamic feature graph to the initial superpixel-based graph. To that purpose, a straightforward manner is to simply average them to produce a refined graph. However, the complementing correlation of the underlying graph structure information cannot be well exploited in this way.

To this end, in this work, an adaptive graph integration module is designed to adaptively exploit the complementary information of the initial superpixel-based graph and the dynamically constructed feature graph. This module can learn proper importance relationships between two graphs. Specifically, after obtaining the dynamic feature graph of the lth graph convolutional layer, the adaptive graph structure in this layer, represented by $\boldsymbol{P}^{(l)}$, can be learned as

$$\boldsymbol{P}^{(l)} = \boldsymbol{s}^{(l)} * \hat{\boldsymbol{A}} + \left(1 - \boldsymbol{s}^{(l)}\right) * \hat{\boldsymbol{A}}_d^{(l)} \tag{6}$$

where $\boldsymbol{s}^{(l)} \in \mathbb{R}^M$ is a score vector controlling the contributions of the initial superpixel-based graph in addition to the dynamically constructed graph, and "$\boldsymbol{u}*\boldsymbol{U}$" means that the ith element of vector $\boldsymbol{u}$ is multiplied with the ith row of the matrix $\boldsymbol{U}$. It should be noted that $\boldsymbol{s}^{(l)}$ is a vector rather a simple scalar. This means that the effects of the original graph and dynamic feature graph are explored at the node level. As a result, all superpixel nodes can adaptively integrate the information from the initial superpixel-based graph and the constructed graphs, and dynamically adjust the contributions from their neighbor nodes. In this work, $\boldsymbol{s}^{(l)}$ can be obtained by

$$\boldsymbol{s}^{(l)} = \phi\left(\boldsymbol{H}^{(l-1)}\boldsymbol{W}_s^{(l)} + \boldsymbol{b}_s^{(l)}\right) \tag{7}$$

where $\phi(\cdot)$ is the sigmoid function, $\boldsymbol{W}_s^{(l)} \in \mathbb{R}^{n\times 1}$ and $\boldsymbol{b}_s^{(l)} \in \mathbb{R}^{M\times 1}$ are learnable parameters, and $\boldsymbol{H}^{(l-1)} = [\boldsymbol{h}_1^{(l)}, \boldsymbol{h}_2^{(l)}, \ldots, \boldsymbol{h}_M^{(l)}] \in \mathbb{R}^{M\times n}$ is the output of the $(l-1)$th layer.

3) DNAGSI: Given a superpixel-based HSI graph $\mathcal{G}$ with M nodes, a graph convolutional layer can be typically formulated as

$$\boldsymbol{H}^{(l)} = \sigma\left(\boldsymbol{P}^{(l)}\boldsymbol{H}^{(l-1)}\boldsymbol{W}^{(l)}\right) \tag{8}$$

where $\boldsymbol{W}^{(l)} \in \mathbb{R}^{n\times n}$ ($l = 1, 2, \ldots, L$) represents weight parameters and $\sigma(\cdot)$ is the rectified linear unit (ReLU) function. Before fed into the network, the input feature $\boldsymbol{X}$ is first forward-propagated through a fully connected layer to obtain the n-dimension feature $\boldsymbol{H}^{(0)}$, i.e., $\boldsymbol{H}^{(0)} = \boldsymbol{X}\boldsymbol{W}^{(0)}$, where $\boldsymbol{W}^{(0)} \in \mathbb{R}^{B\times n}$ is a trainable parameter matrix.

Nevertheless, the performance of the conventional graph convolution in (8) usually degrades as stacking more layers due to the over-smoothing problem arising from aggregating neighbor spatial information excessively. This means that the learned deep features in higher layers become indistinguishable since the original HSI information is missing during the feature aggregation process. For this, inspired by the works of [33] and [39], we seek to introduce a connection to the initial HSI features $\boldsymbol{H}^{(0)}$ and incorporate an identity mapping $\boldsymbol{I}_n$ into the weight matrix $\boldsymbol{W}^{(l)}$ in our model. The former could assure that the learned graph feature representations of all superpixel nodes can retain at least a fraction of initial HSI information to improve the classification performance, whereas the latter could guarantee that the HSI classification accuracy of the proposed DNAGSI is not worse than that of its shallow variant. When numerous graph convolutional layers are used, the discrimination of the learned HSI graph representations improves. In summary, at the lth graph convolutional layer of the proposed DNAGSI, the mathematical expression can be described as

$$\boldsymbol{H}^{(l)} = \sigma\left(\left((1-\alpha)\boldsymbol{P}^{(l)}\boldsymbol{H}^{(l-1)} + \alpha\boldsymbol{H}^{(0)}\right)\left((1-\beta)\boldsymbol{I}_n + \beta\boldsymbol{W}^{(l)}\right)\right) \tag{9}$$

where $\alpha \in [0, 1]$ and $\beta \in [0, 1]$ are two hyperparameters. By this way, the over-smoothing problem can be alleviated significantly.

C. Joint Loss

Benefitting from the initial spatial adjacency matrix $\hat{\boldsymbol{A}}$ and the dynamic adjacency matrix $\hat{\boldsymbol{A}}_d^{(l)}$, the superpixel nodes with near spatial distances and similar features are pushed to become similar. However, when more layers are added, the separability of all superpixel nodes will be damaged. This

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:55:16 UTC from IEEE Xplore. Restrictions apply.

is always undesirable since the nodes with dissimilar initial spectral signatures should keep away from each other.

To address this issue, we adopted the center loss to regularize the learned deep features in order to make similar nodes have similar feature representations while keeping the separability of nodes with different spectral signatures in the learned embedding space. The center loss has been used in CNNs [40]. However, in this article, we incorporate it into the proposed HSI-based graph method, and intuitively aim at encouraging intraclass similarities of hyperspectral pixels. This can be accomplished by reducing the Euclidean distance between each learned feature and its corresponding class center feature. To this end, the learned graph feature representations should be first transformed into the HSI space. The HSI classification intends to classify each hyperspectral pixels instead of the superpixel-based node; therefore, we assign a feature representation to each hyperspectral pixel based on the learned node features. If a hyperspectral pixel belongs to some superpixel v, then the corresponding node feature of v is assigned to this hyperspectral pixel. Once completed the assignment procedure, the learned graph feature representations can be projected back to the HSI space.

Suppose that $\boldsymbol{Z}^*$ is the learned feature representation of HSI and denote $\mathcal{D}_L^* = \{(\boldsymbol{z}_{l_i}^*, y_i)\}_{i=1}^N$ as its labeled samples set. Then the center loss can be formulated as

$$\mathcal{L}_{\text{cl}} = \frac{1}{2|\mathcal{D}_L^*|}\sum_{i=1}^{|\mathcal{D}_L^*|}\left\|\boldsymbol{z}_{l_i}^* - \boldsymbol{c}_{y_i}\right\|^2 \tag{10}$$

where $\boldsymbol{c}_k$ is the center of the kth class obtained by averaging the learned deep graph representations of the kth class ($k = 1, 2, \ldots, C$). It should be emphasized that, in this work, C class centers are updated using all training data, rather than only the mini-batch as in the original center loss [40]. This is due to the fact that the input of DNAGSI is a superpixel-based graph, allowing it to output the graph representations of all pixels directly.

Furthermore, the conventional optimization objective for the HSI classification task is defined as

$$\mathcal{L}_{\text{class}} = \frac{1}{|\mathcal{D}_L^*|}\sum_{(\boldsymbol{z}_{l_i}^*, y_i)\in\mathcal{D}_L^*}\ell\left(\text{softmax}(\boldsymbol{W}^{(L+1)}\boldsymbol{z}_{l_i}^*), y_i\right) \tag{11}$$

where $\boldsymbol{W}^{(L+1)}$ is the weight parameters, softmax$(\cdot)$ represents the softmax function, and $\ell(\cdot,\cdot)$ is the cross entropy loss [41], [42].

Consequently, we build a joint loss to supervise the final classification score, defined as

$$\mathcal{L}_{\text{joint}} = \mathcal{L}_{\text{class}} + \lambda\mathcal{L}_{\text{cl}} \tag{12}$$

where $\lambda > 0$ is a hyperparameter that balances the effects of the cross entropy loss and the center loss. All the parameters of DNAGSI are denoted by $\boldsymbol{W} = \{\boldsymbol{W}^{(0)}, \boldsymbol{W}^{(L+1)}, \{\boldsymbol{W}^{(l)}, \boldsymbol{W}_s^{(l)}, \boldsymbol{b}_s^{(l)}\}_{l=1}^L\}$. Using this new loss, we can cluster the learned graph representations of intraclasses effectively and meanwhile improve the separability of intraclasses and interclasses. To sum up, the entire algorithm for DNAGSI is summarized in **Algorithm 1**.

Algorithm 1 Training Algorithm of DNAGSI

Input: Input graph feature $\boldsymbol{X}$, hyperparameters α, β and λ, learning rate η, maximal iterations $\tau_{\max}$.
Output: $\boldsymbol{W}$.

Set $m = 1$, randomly initialize $\boldsymbol{W}$ and $\boldsymbol{b}$;
repeat
  $\boldsymbol{H}^{(0)} = \boldsymbol{X}\boldsymbol{W}^{(0)}$;
  for $l = 1, 2, \cdots, L$ **do**
    $\boldsymbol{s}^{(l)} = \phi(\boldsymbol{H}^{(l-1)}\boldsymbol{W}_s^{(l)} + \boldsymbol{b}_s^{(l)})$;
    $\boldsymbol{P}^{(l)} = \boldsymbol{s}^{(l)} * \hat{\boldsymbol{A}} + (1 - \boldsymbol{s}^{(l)}) * \hat{\boldsymbol{A}}_d^{(l)}$;
    $\boldsymbol{H}^{(l)} = \sigma(((1-\alpha)\boldsymbol{P}^{(l)}\boldsymbol{H}^{(l-1)} + \alpha\boldsymbol{H}^{(0)})((1-\beta)\boldsymbol{I}_n + \beta\boldsymbol{W}^{(l)}))$;
  end
  Transform $\boldsymbol{H}^{(L)}$ into the HSI space ;
  Calculate $\mathcal{L}_{\text{cl}}(\boldsymbol{W})$ by Eq.(10);
  Compute $\mathcal{L}_{\text{class}}(\boldsymbol{W})$ by Eq.(11);
  $\mathcal{L}_{\text{joint}}(\boldsymbol{W}) \leftarrow \mathcal{L}_{\text{cl}}(\boldsymbol{W}) + \mathcal{L}_{\text{class}}(\boldsymbol{W})$;
  $\boldsymbol{W} \leftarrow \boldsymbol{W} - \eta\frac{\partial\mathcal{L}_{\text{joint}}(\boldsymbol{W})}{\partial\boldsymbol{W}}$
  $m \leftarrow m + 1$;
until $m > \tau_{max}$;

IV. Experiment

In this section, extensive experiments are undertaken to demonstrate the effectiveness of DNAGSI in HSI classification tasks. All experiments are done out in Python environment using Intel Core i7-6700 CPU and GeForce GTX 1060 with 16-GB main memory.

A. Experimental Settings

1) Datasets: Several benchmark HSI datasets, including the Salinas, Indian Pines, and Pavia University datasets, are used in the experiments. Concretely, the Salinas dataset was collected by the airborne visible/infrared imaging spectrometer (AVIRIS) instrument from the Salinas Valley area of California. It has 224 bands and we only use 204 bands for the experiment since 20 bands are polluted by water absorption. Each spectral band is of size 512×214, and the total ground truth includes 16 classes. Furthermore, the Indian Pines dataset was taken in north-western Indiana by AVIRIS in 1992, which comprised of 145×145 pixels with 200 spectral bands after deleting some water absorption bands. There are 16 different types of vegetation. In addition, the Pavia University dataset, containing nine classes of urban land covers and has 103 spectral bands with 610×340 pixels, was acquired by the reflective optical system imaging spectrometer (ROSIS) from northern Italy in 2001.

2) Parameter Setting: In the experiments, μ is set to 80, 25, and 80 for the Salinas, Indian Pines, and Pavia University datasets, respectively. Furthermore, α is, respectively, set as 0.3, 0.3, and 0.4 for the Salinas, Indian Pines, and Pavia University datasets. Similarly, β is set as 0.5 for three datasets empirically for simplicity. Again, set $\lambda = 0.5$ and the number of layers L in DNAGSI is assigned to 6, 6, and 4 for three datasets, respectively. The numbers of hidden units n in DNAGSI are all set as 128. These hyperparameters will be discussed in Section IV-D. In addition, the proposed DNAGSI

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:55:16 UTC from IEEE Xplore. Restrictions apply.

TABLE I
CLASSIFICATION ACCURACIES FOR THE SALINAS DATASET

Class	train	test	CCNN	DFFN	2D-CNN	SSRN	LMAFN	GCN	CEGCN	MDGCN	$\text{DNAGSI}_{\text{base}}$	DNAGSI
1	21	1988	98.75	89.27	99.80	99.57	99.43	99.84	**100.00**	**100.00**	**100.00**	**100.00**
2	38	3688	99.43	99.35	99.79	**100.00**	99.69	99.83	**100.00**	99.26	99.80	**100.00**
3	20	1956	98.04	99.61	98.92	**100.00**	99.87	98.52	**100.00**	99.12	99.41	**100.00**
4	14	1380	98.20	97.57	98.22	**99.46**	99.15	96.25	98.97	97.38	96.24	99.30
5	27	2651	96.68	99.26	98.90	**99.88**	98.43	96.29	97.51	97.05	97.40	97.98
6	40	3919	99.90	99.95	99.96	**100.00**	**100.00**	99.61	99.89	99.56	99.71	99.69
7	36	3543	99.27	99.46	99.77	**99.99**	99.94	99.66	99.90	99.48	99.49	99.97
8	113	11158	83.81	85.12	91.31	94.23	88.16	97.90	98.79	98.61	**99.96**	99.94
9	63	6140	99.31	99.35	99.91	**100.00**	99.94	99.85	**100.00**	99.75	**100.00**	**100.00**
10	33	3245	93.33	94.21	95.75	98.51	97.13	94.40	96.26	94.76	97.73	**99.08**
11	11	1057	94.59	92.99	96.89	99.18	99.15	93.57	99.47	95.81	99.51	**99.87**
12	20	1907	99.22	99.97	99.91	**100.00**	99.94	99.98	98.79	99.63	98.29	99.99
13	10	906	98.74	99.13	98.96	**99.30**	99.26	98.42	93.89	98.16	92.09	98.30
14	11	1059	97.54	96.49	98.13	98.29	**99.15**	97.97	97.80	95.97	94.95	96.95
15	73	7195	70.81	77.95	89.51	90.94	86.76	97.31	99.42	97.92	99.69	**99.99**
16	19	1788	95.18	97.43	97.10	99.60	99.12	96.27	99.55	97.28	**100.00**	**100.00**
OA(%)	-	-	91.44	92.61	96.12	97.38	95.35	98.12	99.06	98.42	99.19	**99.70**
AA(%)	-	-	95.17	95.44	97.68	98.68	97.82	97.85	98.76	98.11	98.39	**99.44**
$\kappa \times 100$	-	-	90.46	91.77	95.68	97.08	94.82	97.91	98.96	98.25	99.10	**99.66**

method is trained by the Adam algorithm [43] for 1000 epochs and the learning rate is given with 0.001. Other baselines adopt the default hyperparameter settings in their implementation. Moreover, the ratios of training samples are, respectively, set as 1%, 10%, and 1% for the Salinas, Indian Pines, and Pavia University datasets.

3) Comparison Methods: The proposed method is compared with numerous state-of-the-art CNN-based and GCN-based HSI classification approaches. Here, the CNN-based methods include the CCNN [17], deep feature fusion network (DFFN) [44], 2D-CNN [16], spectral–spatial residual network (SSRN) [22], and lightweight multiscale attention fusion network (LMAFN) [45]. Furthermore, the GCN-based methods contain GCN [25], CEGCN [30], and multiscale dynamic GCN (MDGCN) [29]. In addition, we compare DNAGSI with its counterpart model, denoted by $\text{DNAGSI}_{\text{base}}$, which has the same basic network architecture but without the adaptive graph structure integration module and the center loss. For a fair comparison, the parameter settings of $\text{DNAGSI}_{\text{base}}$ are the same as those of DNAGSI.

4) Indices: For quantitative evaluation, several commonly used indices, such as overall accuracy (OA), average accuracy (AA), kappa coefficient (κ), and individual class accuracy are adopted [46]. The larger OA, AA, κ, and individual class accuracies, the better the results. All the results are recorded as the average values of ten independent runs.

B. Comparison Results

1) Performance on Salinas Dataset: Table I compares the experimental results of DNAGSI to various baselines on the Salinas dataset. It can be easily found that DNAGSI significantly outperforms other baselines in eight out of sixteen classes. Simultaneously, DNAGSI also displays the best OA, AA, and κ values. These illustrate that our method can adaptively capture informative graph structure and effectively improve the graph representation ability. Concretely, the proposed method outperforms the CNN-based methods. One probable explanation is that our method can exploit long-range spatial contextual information by constructing graph structure in HSI to boost classification accuracy. It is worth mentioning that DNAGSI, which proposes to learn a refined graph structure, provides reasonably good classification accuracies and significantly surpasses GCN that does not consider the dynamic graph structure. This demonstrates the utility of dynamic graph structure learning for HSI classification, and the significance of learning proper graph structures for exploring effective graph representation of HSI. Compared with CEGCN and MDGCN, the proposed method achieves better performance. This is since DNAGSI can produce more reliable and effective graph structures by adaptively integrating the initial graph and the dynamically constructed graphs in each layer, improving the discriminative ability. Besides, DNAGSI is superior to the baseline $\text{DNAGSI}_{\text{base}}$. The possible reason for this is that our method is able to enhance the separability via the adaptive graph structure learning and the center loss. These analyses indicate the superiority of DNAGSI.

From the perspective of visualization, Fig. 2 gives the classification results of various approaches on the Salinas dataset. The visual effects produced by CCNN, DFFN, 2D-CNN, SSRN, and LMAFN are extremely poor. In particular, CCNN, DFFN, and LMAFN suffer from a large number of noisy pixels in two greatest homogeneous regions. When the visual maps of CNN-based methods and GCN-based methods are compared, it is clear that GCN-based methods perform significantly better in different homogeneous regions, both large and small. This validates the potential ability of GCN-based methods to aggregate the spatial information of HSI. Based on the visualizations, it is obvious that DNAGSI produces results that are significantly more similar to the ground truth. Furthermore, DNAGSI performs better than CEGCN and $\text{DNAGSI}_{\text{base}}$ in some small geographical regions, and it also outperforms MDGCN. It is evident that the DNAGSI can fully utilize spatial information by using the underlying graph structure of HSI. In addition, the joint loss enhances the visual result by improving the discrimination of the learned graph representations. In short, these comparisons demonstrate the superior performance of DNAGSI on this dataset.

2) Performance on Indian Pines Dataset: Actually, Table II summarizes the comparative results on the Indian Pines dataset

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:55:16 UTC from IEEE Xplore. Restrictions apply.

TABLE II
CLASSIFICATION ACCURACIES FOR THE INDIAN PINES DATASET

Class	train	test	CCNN	DFFN	2D-CNN	SSRN	LMAFN	GCN	CEGCN	MDGCN	$\text{DNAGSI}_{\text{base}}$	DNAGSI
1	5	41	79.51	73.17	90.98	94.63	**96.83**	94.15	94.82	87.80	93.41	92.68
2	143	1285	93.41	94.80	96.42	97.42	**98.05**	95.42	95.35	96.79	96.53	95.79
3	83	747	94.98	95.84	98.19	97.91	**98.84**	95.14	91.77	96.31	96.95	98.31
4	24	213	87.28	89.44	96.06	**98.50**	95.07	92.58	91.49	93.99	95.45	95.42
5	49	434	94.77	94.33	95.16	96.01	97.10	94.35	95.33	94.72	96.61	**99.71**
6	73	657	97.98	98.81	98.93	99.39	**99.54**	97.70	98.23	98.13	98.87	99.12
7	3	25	81.60	78.40	**98.40**	92.40	92.00	90.00	87.50	87.20	90.00	94.50
8	48	430	99.30	99.44	99.98	99.70	100.00	99.21	99.97	99.40	99.65	**100.00**
9	2	18	62.22	83.89	89.44	85.56	93.33	94.44	87.50	78.33	99.44	**100.00**
10	98	874	90.02	92.51	95.19	97.05	**97.16**	92.81	90.23	93.34	94.60	96.61
11	246	2209	95.86	96.91	98.37	98.29	98.42	98.08	97.91	97.67	98.61	**99.35**
12	60	533	86.36	89.85	97.50	96.72	97.49	96.59	95.05	96.74	97.69	**98.12**
13	21	184	99.40	99.35	**100.00**	99.78	99.67	97.61	98.71	97.55	98.48	97.28
14	127	1138	97.58	97.38	98.80	99.20	99.30	99.52	99.88	99.24	99.73	**100.00**
15	39	347	86.80	90.58	94.12	97.00	94.55	97.84	**98.92**	97.32	98.70	98.88
16	10	83	94.82	97.11	**99.64**	97.47	98.43	97.95	91.57	98.07	97.35	98.80
OA(%)	-	-	94.16	95.35	97.53	98.00	98.20	96.72	96.19	96.93	97.73	**98.36**
AA(%)	-	-	90.12	91.99	96.70	96.69	97.24	95.84	94.64	94.54	97.00	**97.79**
$\kappa \times 100$	-	-	93.34	94.70	97.18	97.72	97.94	96.26	95.65	96.50	97.41	**98.12**

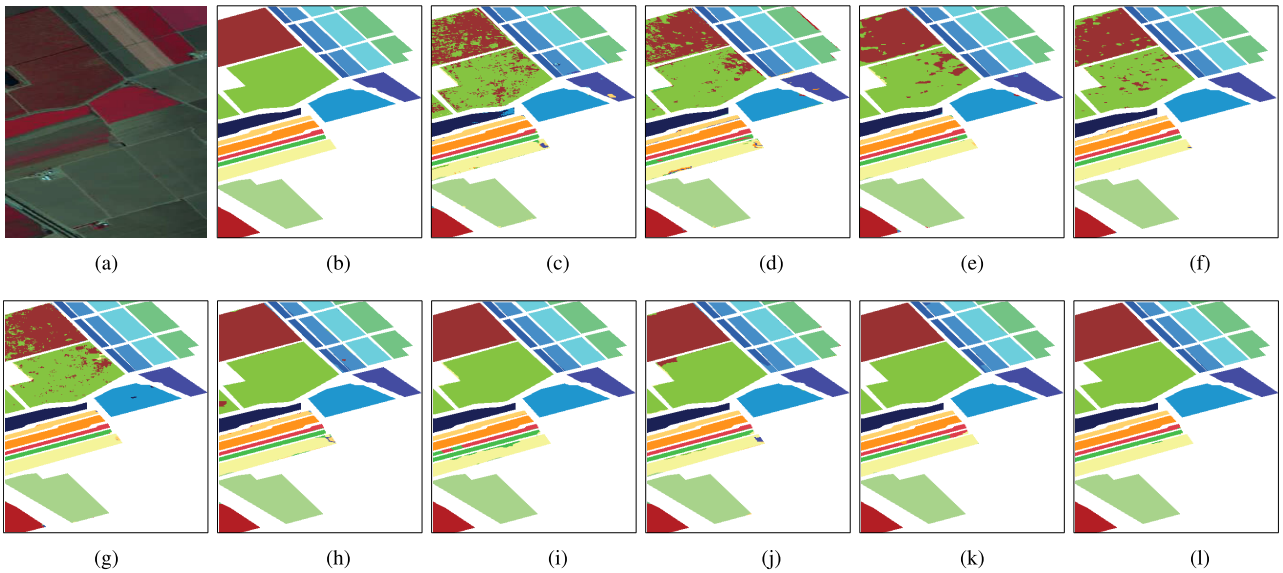


Fig. 2. The classification map for the Salinas dataset. (a) false color image. (b) ground truth. (c) CCNN. (d) DFFN. (e) 2D-CNN. (f) SSRN. (g) LMAFN. (h) GCN. (i) CEGCN. (j) MDGCN. (k) $\text{DNAGSI}_{\text{base}}$. (l) DNAGSI.

and Fig. 3 shows the corresponding classification visual images. We can see from Fig. 3(b) that the Indian Pines dataset contains complex spatial distribution of the land cover; especially, there are some detailed spatial structures. Besides, this dataset is also heavily contaminated by noises. Accordingly, it is challenging to classify the hyperspectral pixels accurately for the Indian Pines dataset. Table II shows that, in addition to SSRN, LMAFN and the proposed DNAGSI, other baselines have lower OAs, implying that plenty of hyperspectral pixels are misclassified and that the produced visual classification maps contain many noisy pixels. But DNAGSI produces the best OA, AA, and κ values among all methods. Especially, DNAGSI outperforms GCN and $\text{DNAGSI}_{\text{base}}$ that without the adaptively learn graph structure. This illustrates the necessity of adopting adaptively dynamic graph structure of HSI in different layers. Moreover, DNAGSI performs better than CEGCN and MDGCN, which validates the advantage of the proposed graph structure learning strategy. The experimental results reveal that DNAGSI can effectively implement the HSI classification on the Indian Pines dataset by making full use of the spatial information through the learned adaptive graph structure. In addition, the comparison of GCN, $\text{DNAGSI}_{\text{base}}$, and DNAGSI illustrates that the initial residual and identity mapping may significantly relieve the oversmoothing problem. Meanwhile, with the center loss and the adaptive graph structure, DNAGSI can further improve the classification accuracy. On the other hand, from the visual classification maps provided in Fig. 3, we can observe many noisy pixels existing in the results of CCNN, 2D-CNN, and GCN, which affect the final results and are consistent with the evaluation metrics in Table II. According to the intuitive comparisons given in Fig. 3, unlike the results in the Salinas dataset, the GCN-based approaches do not clearly outperform the CNN-based methods. But the classification map of DNAGSI is still closest to the ground truth among all methods. This further shows that DNAGSI can dramatically improve feature representation capability for satisfying classification performance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:55:16 UTC from IEEE Xplore. Restrictions apply.

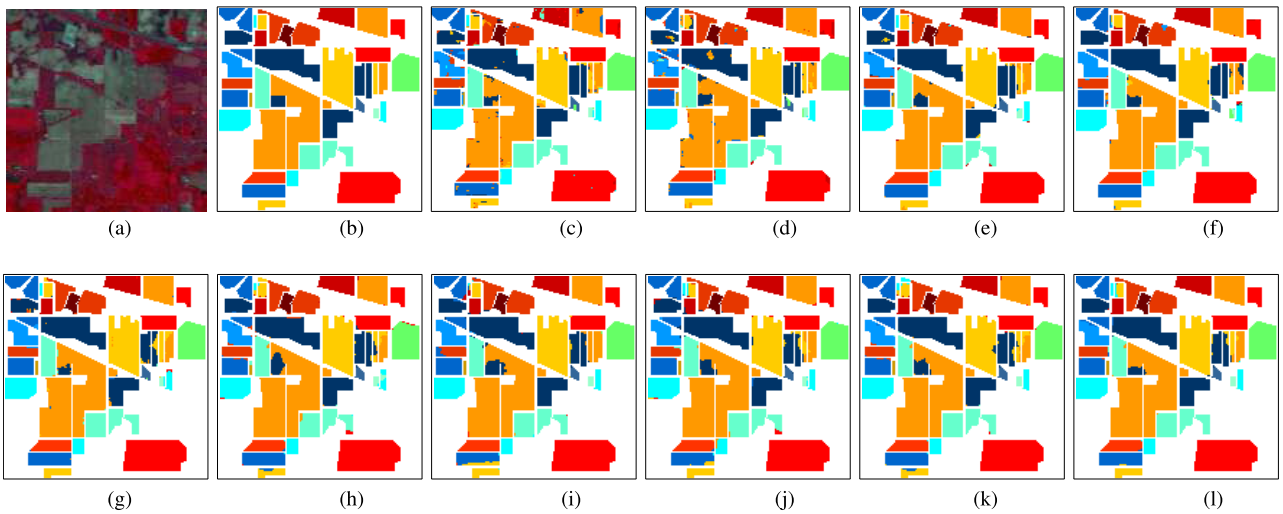


Fig. 3. Classification maps for the Indian Pines dataset. (a) False color image. (b) Ground truth. (c) CCNN. (d) DFFN. (e) 2D-CNN. (f) SSRN. (g) LMAFN. (h) GCN. (i) CEGCN. (j) MDGCN. (k) $\text{DNAGSI}_{\text{base}}$. (l) DNAGSI.

TABLE III
CLASSIFICATION ACCURACIES FOR THE PAVIA UNIVERSITY DATASET

Class	train	test	CCNN	DFFN	2D-CNN	SSRN	LMAFN	GCN	CEGCN	MDGCN	$\text{DNAGSI}_{\text{base}}$	DNAGSI
1	67	6564	91.55	92.65	97.01	98.56	96.51	96.05	94.65	96.12	98.91	**99.72**
2	187	18462	98.75	98.64	99.50	99.46	99.84	99.36	99.59	99.58	99.67	**99.96**
3	21	2078	67.05	79.30	89.03	88.35	90.70	96.87	96.45	94.00	**100.00**	99.81
4	31	3033	92.45	92.77	93.86	**97.85**	95.61	76.86	86.10	86.63	91.89	93.77
5	14	1331	98.44	98.68	99.53	**100.00**	99.80	96.19	99.04	97.91	99.56	99.53
6	51	4978	84.06	90.77	96.22	98.11	96.84	98.71	99.09	97.51	**99.81**	99.76
7	14	1316	68.59	77.63	91.93	98.54	93.66	98.72	99.60	94.98	99.82	**99.83**
8	37	3645	80.72	86.21	95.02	98.77	94.97	96.78	94.51	96.38	97.84	**99.20**
9	10	937	87.20	96.44	99.16	**99.94**	99.52	72.39	76.63	85.50	96.54	97.01
OA(%)	-	-	91.15	93.65	97.18	98.44	97.60	96.10	96.68	96.82	98.80	**99.30**
AA(%)	-	-	85.42	90.34	95.69	97.73	96.38	92.44	93.96	94.29	98.23	**98.73**
$\kappa \times 100$	-	-	88.14	91.54	96.26	97.93	96.82	94.82	95.59	95.77	98.41	**99.07**

3) Performance on Pavia University Dataset: The quantitative results are listed in Table III, and the corresponding visual classification maps on the Pavia University dataset are shown in Fig. 4. It can be observed that CCNN has lower OAs, and this method results in numerous noisy pixels in the generated visual classification map. The visual map of 2D-CNN contains many false labels, which is consistent with the low OA in Table III. GCN also achieves lower classification accuracy. In contrast, CEGCN, MDGCN, and the proposed DNAGSI have higher OAs when compared with GCN. The comparison with existing graph-structure-learning approaches, such as CEGCN and MDGCN, proves the vast superiority of the proposed method again. The reason is that DNAGSI can enhance the initial graph structure and learn a more refined one, which improves the ability to extract graph representations. According to the intuitive comparisons in Fig. 4, GCN-based deep approaches can yield much superior classification maps when compared with CNN-based methods. Among these methods, DNAGSI has the best OA, AA, and κ values and produces the best classification map that is very close to the ground truth. The above-mentioned analyses illustrates that DNAGSI is effective on this dataset and is powerful in discriminative HSI feature extraction.

4) Computational Complexity and Running Time: Table IV gives the number of parameters, training time, and test time for the GCN-based methods. The results are reported in Table IV. It can be seen that DNAGSI, CEGCN, and MDGCN spend more time during the training stage, while GCN takes lesser for training. This is since DNAGSI, CEGCN, and MDGCN require extra time to obtain the dynamic graph in each layer, resulting in more training time. But DNAGSI does not significantly increase the training time as the number of network layers increases compared with GCN. Meanwhile, DNAGSI shows higher computational efficiency than CEGCN and MDGCN during the training phase on the Salinas and Pavia University datasets. The main reason is that DNAGSI only requires a smaller number of floating-point operations to learn the dynamic graph structure in comparison with CEGCN and MDGCN. Moreover, the number of parameters for DNAGSI is higher than those of other methods. It is reasonable due to the fact that DNAGSI is a deep framework, whereas the other methods are shallow models.

5) Various Training Percentage: We discuss the influence of different percent of training samples for the proposed method on three datasets. Concretely, for the Indian Pines dataset, we randomly choose 5%, 10%, 15%, and 20% of the labeled samples as the training set and the remaining labeled samples are used for testing. For the Pavia University and Salinas datasets, 0.5%, 1%, 2%, and 5% of the labeled samples are randomly selected for training and the rest are used for testing. The results are listed in Table V. As can be seen, the classification performance of DNAGSI progressively

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:55:16 UTC from IEEE Xplore. Restrictions apply.

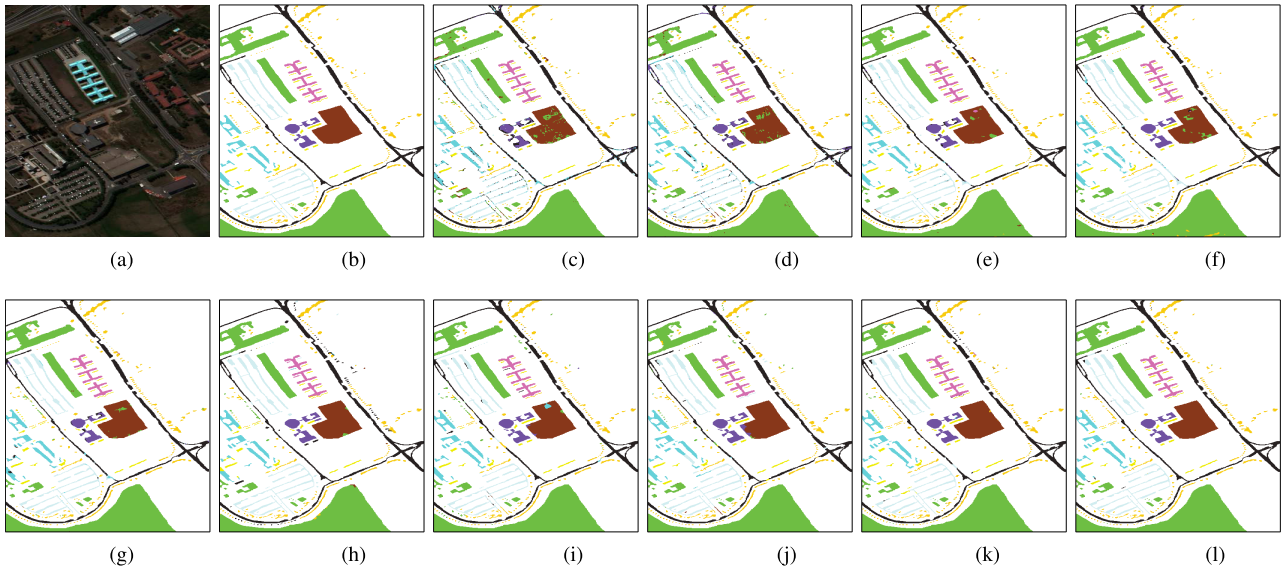


Fig. 4. Classification maps for the Pavia University dataset. (a) False color image. (b) Ground truth. (c) CCNN. (d) DFFN. (e) 2D-CNN. (f) SSRN. (g) LMAFN. (h) GCN. (i) CEGCN. (j) MDGCN. (k) $\text{DNAGSI}_{\text{base}}$. (l) DNAGSI.

TABLE IV
NUMBER OF PARAMETERS, TRAINING TIME, AND TEST TIME ON THREE DATASETS

Methods	Salinas			Indian Pines			Pavia University		
	Train time	Test time	Parameters	Train time	Test time	Parameters	Train time	Test time	Parameters
GCN	58.8	0.008	45,480	9.6	0.003	44,960	2.48	0.007	31,447
CEGCN	104.4	0.009	130,984	23.4	0.006	128,784	5.36	0.011	91,095
MDGCN	163.8	0.01	84,912	67.2	0.017	83,376	9.18	0.019	43,419
$\text{DNAGSI}_{\text{base}}$	63	0.016	110,864	23.4	0.005	110,352	2.67	0.011	64,009
DNAGSI	96	0.066	194,145	49.8	0.022	193,629	3.12	0.052	113,907

TABLE V
OA (%) OF DNAGSI WITH DIFFERENT TRAINING PERCENTAGE

Salinas	Training percentage	0.5%	1%	2%	5%
	OA	98.75	99.70	99.78	99.84
Indian Pines	Training percentage	5%	10%	15%	20%
	OA	97.49	98.36	98.63	98.98
Pavia University	Training percentage	0.5%	1%	2%	5%
	OA	98.79	99.30	99.43	99.73

TABLE VI
INFLUENCE OF DYNAMIC GRAPH STRUCTURE LEARNING

OA(%)	Salinas	Indian Pines	Pavia University
GCN	98.12	96.72	96.10
GCN_{agsi}	99.08	97.65	97.02
$\text{DNAGSI}_{\text{no}}$	99.25	97.82	98.88
DNAGSI	99.70	98.36	99.30

TABLE VII
IMPACT OF ADAPTIVE GRAPH INTEGRATION

OA(%)	Salinas	Indian Pines	Pavia University
$\text{DNAGSI}_{\text{no}}$	99.25	97.82	98.88
$\text{DNAGSI}_{\text{avg}}$	99.48	98.04	98.95
DNAGSI	99.70	98.36	99.30

improves with the increased numbers of training samples. In particular, DNAGSI produces excellent performance even when the number of training samples is rather limited.

To sum up, the proposed DNAGSI approach yields superior performance over classical models on three real HSI datasets.

TABLE VIII
EFFECT OF INITIAL RESIDUAL AND IDENTITY MAPPING

OA(%)	Salinas	Indian Pines	Pavia University
DNAGSI_{*}	97.83	95.82	95.65
DNAGSI	99.70	98.36	99.30

TABLE IX
ROLE OF CENTER LOSS IN JOINT LOSS

OA(%)	Salinas	Indian Pines	Pavia University
$\text{DNAGSI}_{\#}$	99.63	98.23	99.18
DNAGSI	99.70	98.36	99.30

C. Ablation Studies

To evaluate the performance of major components of DNAGSI, the ablation studies on HSI datasets are conducted.

1) Influence of Dynamic Graph Structure Learning: First, we design one variation, denoted by $\text{DNAGSI}_{\text{no}}$, which only uses the initial fixed graph during the graph representation learning. Besides, we also utilize GCN as the baseline framework to further verify its effectiveness. Specifically, GCN_{agsi} is designed to represent the model that exploits the proposed dynamic graph structure learning component in graph convolutional layer. The results are given in Table VI. Obviously, DNAGSI and GCN_{agsi} significantly outperform the corresponding baseline models $\text{DNAGSI}_{\text{no}}$ and GCN, verifying that the dynamic graph structure learning can effectively represent the graph structure in different layers and improve the feature extraction capability.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:55:16 UTC from IEEE Xplore. Restrictions apply.

TABLE X
INFLUENCE OF α ON THREE DATASETS

α	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
Salinas	98.28	99.62	99.63	**99.70**	99.67	99.50	99.40	99.26	99.16	98.91	95.31
Indian Pines	96.62	98.13	98.26	**98.36**	98.34	98.12	98.06	97.77	98.06	97.88	95.32
Pavia University	97.11	98.08	98.34	99.15	**99.30**	99.28	99.22	98.98	98.93	98.90	95.49

TABLE XI
INFLUENCE OF β ON THREE DATASETS

β	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
Salinas	99.42	99.47	99.47	99.56	99.65	**99.70**	99.64	99.59	99.48	99.38	99.06
Indian Pines	97.72	97.83	98.05	98.14	98.23	**98.36**	98.33	98.31	98.13	97.78	97.38
Pavia University	98.64	99.15	99.23	99.19	**99.31**	99.30	**99.31**	99.30	99.25	99.08	99.10

2) Effect of Adaptive Graph Structure Integration: Let $\text{DNAGSI}_{\text{avg}}$ be the model which integrates the initial graph structure and the learned dynamic graph structure by a simple average operation. $\text{DNAGSI}_{\text{no}}$ is also used as the baseline. The related results are recorded in Table VII. It can be clearly observed that DNAGSI and $\text{DNAGSI}_{\text{avg}}$ perform better than the $\text{DNAGSI}_{\text{no}}$. This illustrates that the learned graph structures in different layers can provide additional information for graph representation learning. More importantly, DNAGSI can further promote the classification performance of $\text{DNAGSI}_{\text{avg}}$. The possible reason is that the proposed adaptive graph structure integration strategy can effectively fuse the initial graph and the learned dynamic graph by adaptive weights and thereby capture the graph structure information of HSI better.

3) Impact of Initial Residual and Identity Mapping: We devise a variant, denoted by DNAGSI_{*}, in which the initial residual and identity mapping are not utilized. Table VIII summaries the classification results on three HSI datasets. As can be seen, DNAGSI can benefit from considerably deeper network architecture, but DNAGSI_{*} only displays lower accuracies. This validates that the initial residual and identity mapping can assist the network alleviate the over-smoothing problem and improving model performance. In addition, the classification accuracies of DNAGSI_{*} are even worse than those of GCN, emphasizing the importance of the initial residual and the identity mapping again.

4) Role of Center Loss in Joint Loss: We use $\text{DNAGSI}_{\#}$ to denote the variant without the center loss to supervise the training process. The results are reported in Table IX. It can be seen that the classification accuracies of the proposed method are better than $\text{DNAGSI}_{\#}$. A possible reason is that the proposed method without the center loss is prone to over-smoothing throughout the training phase. This reveals that the center loss can force the learned graph representations of HSI to be significantly more discriminative.

D. Hyperparameter Analyses

In this section, some hyperparameters like α, β, λ, the number of layers L, and the number of hidden units n are discussed.

In reality, α is a key factor of DNAGSI, which controls the percentage of the original HSI information that is utilized by the hidden layers to learn graph representations. Usually, the range is in the interval [0, 1]. Table X shows the influences of different α on the classification accuracies of the proposed DNAGSI method for the Salinas, Indian Pines and Pavia University datasets. It is easy to see that the OAs first grow and then gradually decrease as α increases. Note that when α is large, e.g., 1.0, the classification accuracy greatly reduced. This suggests that the original HSI information is overused. As a result, the potential features cannot be effectively transmitted to the subsequent layers. Similarly, the performance decreases if $\alpha = 0$, proving the importance of initial graph feature. Overall, we chose $\alpha = 0.3$ for the first two datasets and $\alpha = 0.4$ for the Pavia University dataset empirically.

Second, we evaluate the influence of β. In fact, we know $\beta \in [0, 1]$. Therefore, we empirically use the grid search method to find the optimum hyperparameter β from $\{0, 0.1, 0.2, \ldots, 1.0\}$ for three datasets. The experimental results are displayed in Table XI. One can see that the optimal value is set as $\beta = 0.5$ for the Salinas and Indian Pines datasets, and $\beta = 0.4$ or 0.6 for the Pavia University dataset. Due to the fact that the result of $\beta = 0.5$ is nearly equal to that of $\beta = 0.4$ for the Pavia University dataset according to Table XI, we also select $\beta = 0.5$ in all the experiments for this dataset, which is consistent with other two datasets. This reveals that identity mapping can help to alleviate the over-smoothing problem to some extent.

Third, the hyperparameter $\lambda > 0$ in the proposed joint loss balances the center loss and the cross entropy loss. We assess its influence, where the search range is empirically constrained in $\{0.1, 0.2, \ldots, 1.0\}$. The results are listed in Table XII. It can be seen that, with the increase of λ, the OAs first increase to the optimal accuracies and then fluctuate around the relatively satisfying results. The best λ is chosen as 0.5 for three datasets. This demonstrates that the center loss can reliably enhance the separability of intraclasses and interclasses.

Next, we analyze the impact of L. Table XIII summaries the classification results on three HSI datasets. It is obvious to find that DNAGSI can still obtain higher accuracies when the network architecture becomes relatively deeper. Moreover, although the model performance slightly decreases as the network depth increases, its corresponding accuracies still demonstrate the superiority over GCN. This indicates that the proposed method can assist to alleviate the over-smoothing problem when extracting the graph features from HSI to some extent. To sum up, we empirically give the value L in DNAGSI with 6, 6, and 4 for the three HSI datasets, respectively.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:55:16 UTC from IEEE Xplore. Restrictions apply.

TABLE XII
INFLUENCE OF λ ON THREE DATASETS

λ	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
Salinas	99.51	99.53	99.60	99.68	**99.70**	99.70	99.63	99.66	99.67	99.65
Indian Pines	98.05	98.21	98.27	98.27	**98.36**	98.31	98.27	98.23	98.16	98.18
Pavia University	99.09	99.15	99.21	99.26	**99.30**	99.29	99.27	99.25	99.19	99.21

TABLE XIII
IMPACT OF L ON THREE DATASETS

L	3	4	5	6	7	8	9	10
Salinas	99.52	99.60	99.66	**99.70**	99.68	99.69	99.68	99.67
Indian Pines	97.94	98.16	98.12	**98.36**	98.27	98.25	98.19	98.13
Pavia University	99.23	**99.30**	99.17	99.20	99.11	99.06	99.04	99.06

TABLE XIV
EFFECT OF n ON THREE DATASETS

n	16	32	64	128	256
Salinas	98.89	99.00	99.57	**99.70**	99.68
Indian Pines	96.43	97.90	98.08	**98.36**	98.28
Pavia University	98.57	99.21	99.25	**99.30**	99.23

Finally, the effect of n for DNAGSI is investigated. The classification results on three datasets are listed in Table XIV. We can see that the OAs remarkably increase and achieve the best in the case of 128 for all the datasets, implying that DNAGSI can benefit from considerably larger number of hidden units. This is reasonable since the model with more hidden units can capture more abundant features from HSI data. In the experiments, we select $n = 128$ empirically.

V. CONCLUSION

This article presents a DNAGSI for HSI classification. First, through adaptively integrating the initial graph and the dynamic graph obtained in each layer, the DNAGSI can capture the refined graph structure to boost the robust graph representation learning of HSI. Furthermore, the provided initial residual and identity mapping can significantly improve the network depth to generate high-level deep graph representations. Again, the center loss embedded into the joint loss can regularize the learned deep graph features and enhance the discriminative ability. When integrating the center loss, initial residual, and identity mapping, DNAGSI can successfully alleviate the over-smoothing problem in HSI graph representation learning and effectively improve the classification performance. Extensive experiment results demonstrate that DNAGSI can achieve state-of-the-art classification performance in terms of both visual classification map and quantitative metrics.

Nevertheless, there still exists some room to improve this framework. First, some hyperparameters, such as α, β, and λ are handcrafted in this article. Accordingly, it is crucial to explore a feasible and adaptive hyperparameter selection strategy. In this aspect, meta learning [47] is likely to be a promising method. Moreover, the proposed method adopts the cosine similarity to construct the dynamic graph in each layer. Although it has acquired better performance for HSI classification, we can also build an adaptive interaction way among different superpixel nodes by the attention mechanism [48] to obtain better dynamic graphs and further promote the HSI classification performance.

REFERENCES

[1] S. Jia *et al.*, "A lightweight convolutional neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4150–4163, May 2021.

[2] J. Peng, Q. Xie, Q. Zhao, Y. Wang, L. Yee, and D. Meng, "Enhanced 3DTV regularization and its applications on HSI denoising and compressed sensing," *IEEE Trans. Image Process.*, vol. 29, pp. 7889–7903, 2020.

[3] J. Peng, Y. Zhou, W. Sun, Q. Du, and L. Xia, "Self-paced nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1501–1515, Feb. 2021.

[4] H. Ye, H. Li, B. Yang, F. Cao, and Y. Tang, "A novel rank approximation method for mixture noise removal of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4457–4469, Jul. 2019.

[5] W. Sun, J. Peng, G. Yang, and Q. Du, "Fast and latent low-rank subspace clustering for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 3906–3915, Jun. 2020.

[6] F. Luo, B. Du, L. Zhang, L. Zhang, and D. Tao, "Feature learning using spatial–spectral Hypergraph discriminant analysis for hyperspectral image," *IEEE Trans. Cybern.*, vol. 49, no. 7, pp. 2406–2419, Jul. 2019.

[7] J. Peng, Y. Zhou, and C. L. P. Chen, "Region-kernel-based support vector machines for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 9, pp. 4810–4824, Sep. 2015.

[8] Q. Wang, J. Lin, and Y. Yuan, "Salient band selection for hyperspectral image classification via manifold ranking," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 6, pp. 1279–1289, Jun. 2016.

[9] Y. Zhou, J. Peng, and C. L. P. Chen, "Dimension reduction using spatial and spectral regularized local discriminant embedding for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 2, pp. 1082–1095, Feb. 2015.

[10] X. Kang, X. Xiang, S. Li, and J. A. Benediktsson, "PCA-based edge-preserving features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 12, pp. 7140–7151, Dec. 2017.

[11] Z. Ye, J. Chen, H. Li, Y. Wei, G. Xiao, and J. A. Benediktsson, "Supervised functional data discriminant analysis for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 2, pp. 841–851, Feb. 2020.

[12] S. Ji, W. Xu, M. Yang, and K. Yu, "3D convolutional neural networks for human action recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 1, pp. 221–231, Jan. 2013.

[13] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and P. S. Yu, "A comprehensive survey on graph neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 4–24, Jan. 2021.

[14] S. Shao *et al.*, "MHFC: Multi-head feature collaboration for few-shot learning," in *Proc. 29th ACM Int. Conf. Multimedia*, 2021, pp. 4193–4201.

[15] Z. Li, H. Huang, Y. Duan, and G. Shi, "DLPNet: A deep manifold network for feature extraction of hyperspectral imagery," *Neural Netw.*, vol. 129, pp. 7–18, Sep. 2020.

[16] S. Yu, S. Jia, and C. Xu, "Convolutional neural networks for hyperspectral image classification," *Neurocomputing*, vol. 219, pp. 88–98, Jan. 2017.

[17] H. Lee and H. Kwon, "Going deeper with contextual CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 26, no. 10, pp. 4843–4855, Oct. 2017.

[18] X. Cao, F. Zhou, L. Xu, D. Meng, Z. Xu, and J. Paisley, "Hyperspectral image classification with Markov random fields and a convolutional neural network," *IEEE Trans. Image Process.*, vol. 27, no. 5, pp. 2354–2367, May 2018.

[19] X. Tang *et al.*, "Hyperspectral image classification based on 3-D octave convolution with spatial–spectral attention network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2430–2447, Mar. 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:55:16 UTC from IEEE Xplore. Restrictions apply.

[20] H. Sun, X. Zheng, and X. Lu, "A supervised segmentation network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 30, pp. 2810–2825, 2021.
[21] H. Gao, Y. Yang, C. Li, L. Gao, and B. Zhang, "Multiscale residual network with mixed depthwise convolution for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3396–3408, Apr. 2021.
[22] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral-spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.
[23] W. Guo, H. Ye, and F. Cao, "Feature-grouped network with spectral–spatial connected attention for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5500413.
[24] L. Mou, X. Lu, X. Li, and X. X. Zhu, "Nonlocal graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8246–8257, Dec. 2020.
[25] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.
[26] A. Sha, B. Wang, X. Wu, and L. Zhang, "Semisupervised classification for hyperspectral images using graph attention networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 1, pp. 157–161, Jan. 2021.
[27] A. Qin, Z. Shang, J. Tian, Y. Wang, T. Zhang, and Y. Y. Tang, "Spectral–spatial graph convolutional networks for semisupervised hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 241–245, Feb. 2019.
[28] X. He, Y. Chen, and P. Ghamisi, "Dual graph convolutional network for hyperspectral image classification with limited training samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5502418.
[29] S. Wan, C. Gong, P. Zhong, B. Du, L. Zhang, and J. Yang, "Multiscale dynamic graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3162–3177, May 2020.
[30] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "CNN-enhanced graph convolutional network with pixel- and superpixel-level feature fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8657–8671, Oct. 2021.
[31] S. Wan, C. Gong, P. Zhong, S. Pan, G. Li, and J. Yang, "Hyperspectral image classification with context-aware dynamic graph convolutional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 597–612, Jan. 2021.
[32] Y. Ding, Y. Guo, Y. Chong, S. Pan, and J. Feng, "Global consistent graph convolutional network for hyperspectral image classification," *IEEE Trans. Instrum. Meas.*, vol. 70, 2021, Art. no. 5501516.
[33] M. Chen, Z. Wei, Z. Huang, B. Ding, and Y. Li, "Simple and deep graph convolutional networks," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1725–1735.
[34] Z. Zhang, P. Cui, and W. Zhu, "Deep learning on graphs: A survey," *IEEE Trans. Knowl. Data Eng.*, vol. 34, no. 1, pp. 249–270, Jan. 2022.
[35] J. Zhou *et al.*, "Graph neural networks: A review of methods and applications," *AI Open*, vol. 1, pp. 57–81, Jan. 2020.
[36] Y. G. Wang, M. Li, Z. Ma, G. Montufar, X. Zhuang, and Y. Fan, "Haar graph pooling," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 9952–9962.
[37] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. Int. Conf. Learn. Represent.*, Apr. 2017.
[38] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.
[39] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 770–778.
[40] Y. Wen, K. Zhang, Z. Li, and Y. Qiao, "A discriminative feature learning approach for deep face recognition," in *Proc. Eur. Conf. Comput. Vis.*, Oct. 2016, pp. 499–515.
[41] S. Shao, L. Xing, R. Xu, W. Liu, Y.-J. Wang, and B.-D. Liu, "MDFM: Multi-decision fusing model for few-shot learning," *IEEE Trans. Circuits Syst. Video Technol.*, early access, Dec. 16, 2021, doi: 10.1109/TCSVT.2021.3135023.
[42] Y. Xu, B. Du, and L. Zhang, "Self-attention context network: Addressing the threat of adversarial attacks for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 30, pp. 8671–8685, 2021.
[43] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent.*, May 2015.
[44] W. Song, S. Li, L. Fang, and T. Lu, "Hyperspectral image classification with deep feature fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3173–3184, Jun. 2018.
[45] J. Wang *et al.*, "NAS-guided lightweight multiscale attention fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8754–8767, Oct. 2021.
[46] H. Sun, X. Zheng, X. Lu, and S. Wu, "Spectral-spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3232–3245, May 2020.
[47] T. M. Hospedales, A. Antoniou, P. Micaelli, and A. J. Storkey, "Meta-learning in neural networks: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, early access, May 11, 2021, doi: 10.1109/TPAMI.2021.3079209.
[48] J. B. Lee, R. A. Rossi, S. Kim, N. K. Ahmed, and E. Koh, "Attention models in graphs: A survey," *ACM Trans. Knowl. Discov. Data*, vol. 13, no. 6, pp. 1–25, Nov. 2019.



Bing Yang received the B.Sc. degree in information and computing science from Henan University, Kaifeng, China, in 2015, and the Ph.D. degree in statistics from the Huazhong University of Science and Technology, Wuhan, China, in 2021.

He is currently a Lecturer with the College of Sciences, China Jiliang University, Hangzhou, China. His research interests include deep learning and image processing.



Feilong Cao received the B.Sc. and M.Sc. degrees in applied mathematics from Ningxia University, Yinchuan, China, in 1987 and 1998, respectively, and the Ph.D. degree in applied mathematics from Xi'an Jiaotong University, Xi'an, China, in 2003.

He was a Research Fellow with the Center of Basic Sciences, Xi'an Jiaotong University, from 2003 to 2004, where he was a Post-Doctoral Research Fellow with the School of Aerospace, from 2004 to 2006. In 2011, he was a Visiting Professor with the Department of Computer Sciences, Chonbuk National University, South Korea, and the Department of Computer Sciences and Computer Engineering, LaTrobe University, Melbourne, Australia, from 2013 to 2014. He is currently a Professor and the Dean of the College of Sciences, China Jiliang University, Hangzhou, China. He has authored or coauthored over 230 scientific papers in refereed journals. His research interests include pattern recognition, neural networks, and approximation theory.



Hailiang Ye received the B.Sc. and M.Sc. degrees in applied mathematics from China Jiliang University, Hangzhou, China, in 2012 and 2015, respectively, and the Ph.D. degree in computational mathematics from the Huazhong University of Science and Technology, Wuhan, China, in 2019.

He is currently a Lecturer with the College of Sciences, China Jiliang University, Hangzhou, China. He has authored or coauthored over 20 scientific papers in refereed journals. His research interests include deep learning, image processing, and point cloud analysis.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:55:16 UTC from IEEE Xplore. Restrictions apply.